

CLI Usage

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1. Using CLI for Prediction

CLI (Command-Line Interface) is a way for users to interact with computers or software programs, where users run programs or execute tasks by typing text commands instead of clicking icons or buttons through a graphical user interface (GUI).

2. CLI Usage Syntax

```
yolo TASK MODE ARGS
```

where TASK (optional) is one of [detect, segment, classify, pose, obb]
MODE (required) is one of [train, val, predict, export, track, benchmark]
ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that override defaults.

3. Image Prediction

Use yolo26n.pt to predict the built-in images in the ultralytics26 project: If the system cannot find the corresponding model file in the directory where the command is run, it will be automatically downloaded (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/pi/ultralytics26/
```

Use yolo26n.pt to detect images in the target folder and output results:

```
yolo predict model=yolo26n.pt source='/home/pi/ultralytics26/assets'
```

```

pi@raspberrypi:~ $ cd /home/pi/ultralytics26/
pi@raspberrypi:~/ultralytics26 $ yolo predict model=yolo26n.pt source='/home/pi/ultralytics26/assets'
WARNING ⚠ Ultralytics settings reset to default values. This may be due to a possible problem with your settings or a recent ultralytics package update.
View Ultralytics Settings with 'yolo settings' or at '/home/pi/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/guides/quickstart/#ultralytics-settings.
Ultralytics 8.4.21 Python-3.11.2 torch-2.1.2 CPU (Cortex-A76)
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFL0Ps

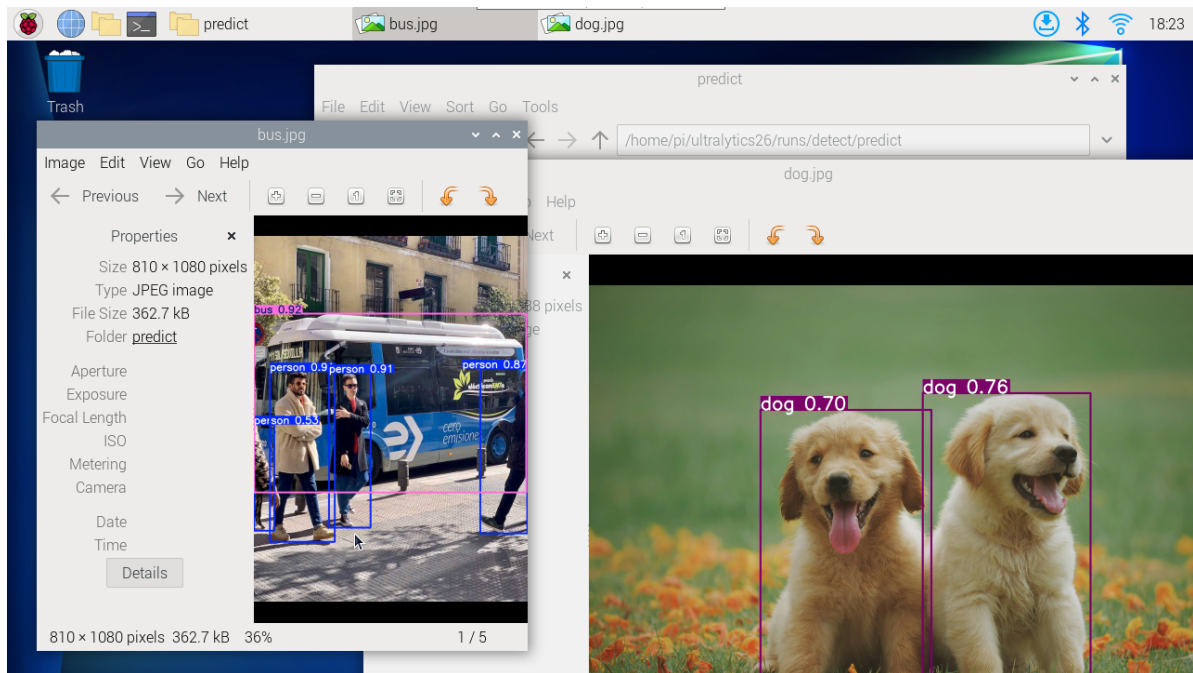
image 1/5 /home/pi/ultralytics26/assets/bus.jpg: 640x480 4 persons, 1 bus, 644.7ms
image 2/5 /home/pi/ultralytics26/assets/car.jpg: 640x640 1 bus, 828.6ms
image 3/5 /home/pi/ultralytics26/assets/dog.jpg: 480x640 2 dogs, 597.8ms
image 4/5 /home/pi/ultralytics26/assets/people.jpg: 448x640 2 persons, 1 bench, 565.8ms
image 5/5 /home/pi/ultralytics26/assets/zidane.jpg: 384x640 2 persons, 1 tie, 492.5ms
Speed: 9.6ms preprocess, 625.9ms inference, 0.8ms postprocess per image at shape (1, 3, 384, 640)
Results saved to /home/pi/ultralytics26/runs/detect/predict
Learn more at https://docs.ultralytics.com/modes/predict
pi@raspberrypi:~/ultralytics26 $

```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/runs/detect/predict

Each time it runs, the predict under this folder automatically increments by 1



4. Video Prediction

Use yolo26n.pt to predict videos in the ultralytics project (not videos that come with ultralytics): If the system cannot find the corresponding model file in the directory where the command is run, it will be automatically downloaded (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/pi/ultralytics26
```

Use yolo26n.pt to detect videos in the target folder and output results:

```
yolo predict model=yolo26n.pt source='/home/pi/ultralytics26/videos'
```

```

pi@raspberrypi:~ $ cd /home/pi/ultralytics26
pi@raspberrypi:~/ultralytics26 $ yolo predict model=yolo26n.pt source='/home/pi/ultralytics26/videos'
Ultralytics 8.4.21 Python-3.11.2 torch-2.1.2 CPU (Cortex-A76)
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFLOPs

video 1/5 (frame 1/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 12 cars, 1 bus, 477.9ms
video 1/5 (frame 2/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 15 cars, 1 bus, 463.5ms
video 1/5 (frame 3/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 13 cars, 1 bus, 445.5ms
video 1/5 (frame 4/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 14 cars, 1 bus, 451.6ms
video 1/5 (frame 5/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 15 cars, 1 bus, 453.6ms
video 1/5 (frame 6/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 13 cars, 446.7ms
video 1/5 (frame 7/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 14 cars, 446.2ms
video 1/5 (frame 8/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 13 cars, 452.7ms
video 1/5 (frame 9/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 17 cars, 445.3ms
video 1/5 (frame 10/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 17 cars, 449.6ms
video 1/5 (frame 11/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 15 cars, 449.5ms
video 1/5 (frame 12/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 16 cars, 453.5ms
video 1/5 (frame 13/180) /home/pi/ultralytics26/videos/car.mp4: 384x640 14 cars, 458.9ms

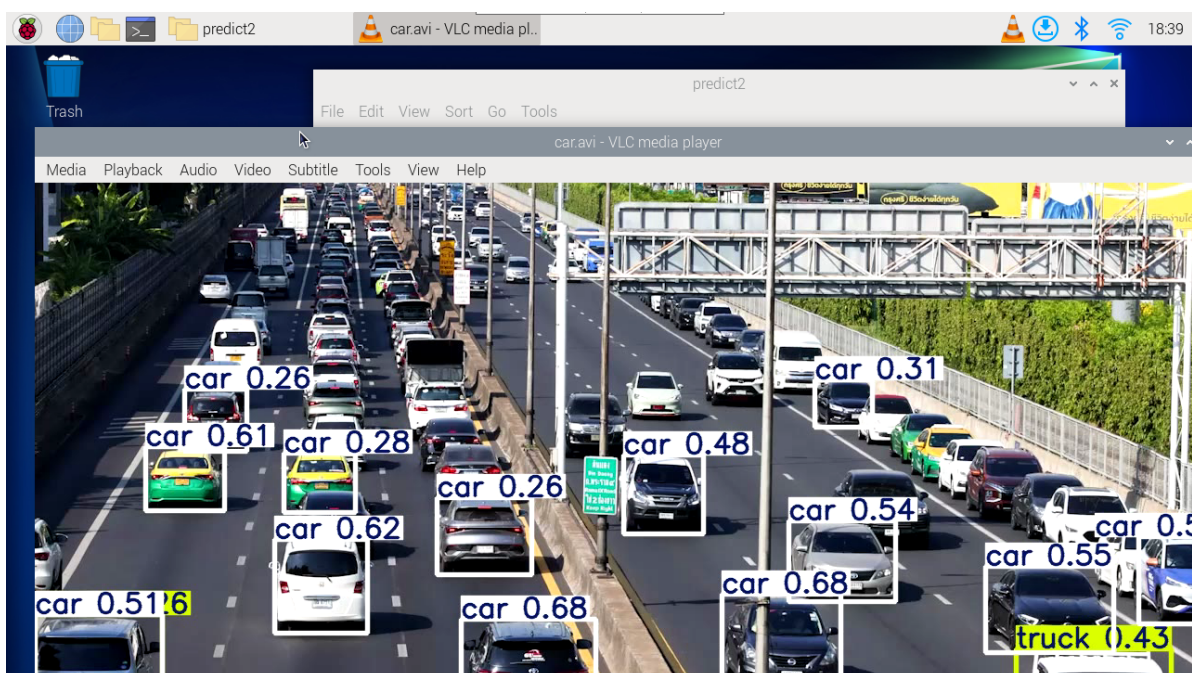
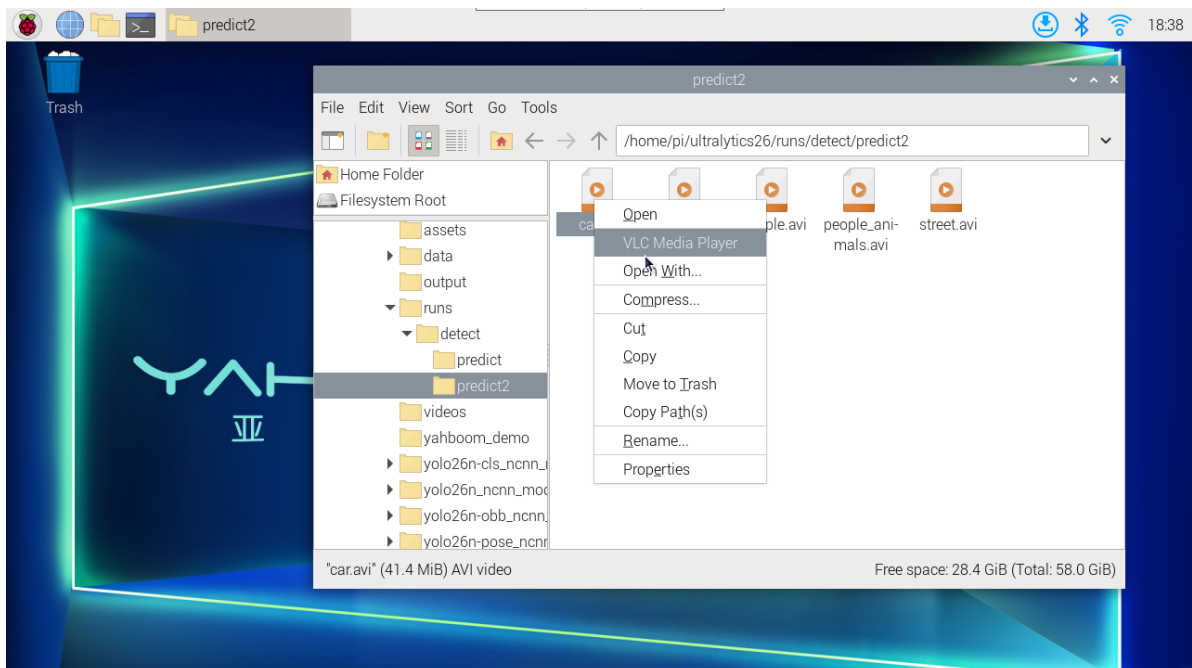
```

Effect Preview

yolo recognition output video location: /home/pi/ultralytics26/runs/detect/predict2

Each time it runs, the predict under this folder automatically increments by 1

You can use the VLC application pre-installed on Raspberry Pi to view videos



5. Real-time Prediction

Use yolo26n.pt to predict USB camera footage: If the system cannot find the corresponding model file in the directory where the command is run, it will be automatically downloaded (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/pi/ultralytics26
```

Use yolo26n.pt to detect camera footage and output results: Currently only USB cameras can directly use CLI to predict real-time footage. No relevant information found for directly using CSI cameras as input source

```
yolo predict model=yolo26n.pt source=0 save=False show # Object detection
# Instance segmentation: yolo predict model=yolo26n-seg.pt source=0 save=False
show
# Image classification: yolo predict model=yolo26n-cls.pt source=0 save=False
show
# Pose estimation: yolo predict model=yolo26n-pose.pt source=0 save=False show
# Oriented object detection: yolo predict model=yolo26n-obb.pt source=0
save=False show
```

Click on the terminal and press the "Ctrl + C" shortcut to terminate the program!

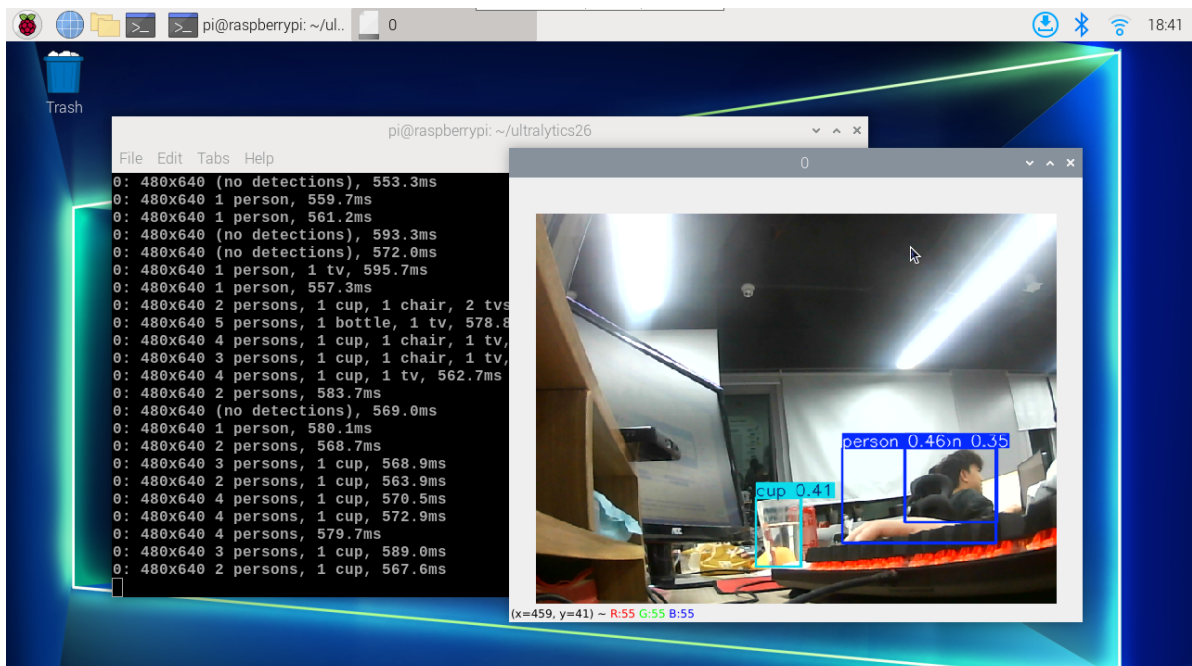
Parameter Description

model : Specify the YOLO model

source : Specify the recognition source: multiple cameras can switch numbers

save=False : Disable saving results

show : Real-time display



References

[Home - Ultralytics YOLO Documentation](#)

[Command Line Interface - Ultralytics YOLO Documentation](#)

